Arman Javan Sekhavat Pishkhani

TEHRAN, IRAN

arman.j.sekhavat@ut.ac.ir

linkedin.com/in/arman-javan-sekhavat

arman-javan-sekhavat.github.io

EDUCATION

Master's Degree, Mechatronics Engineering **2024–2027**
University of Tehran TEHRAN, IRAN

- GPA: 18.43/20 (Latest)
- Thesis Title: Simulation and Intelligent Control of an Endotracheal Intubation Robot
- Thesis Grade: -
- Supervisor: Dr. Khalil Alipour

Bachelor's Degree, Mechanical Engineering **2020–2024**
K. N. Toosi University of Technology TEHRAN, IRAN

- GPA: 3.79/4 (17.95/20) – 3.81/4 (17.99/20) Last two years
- Thesis Title: Design of a Controller for a Robotic Bicycle Using Intelligent Control Techniques
- Thesis Grade: A (19.00/20)
- Supervisor: Dr. S. Hossein Sadati

High School Diploma, STEM **2017–2020**
National Organization for Development of Exceptional Talents (SAMPAD) RASHT, GUILAN

RESEARCH INTERESTS

- Robotics
- Reinforcement Learning
- Control Theory

TEACHING EXPERIENCE

Teaching Assistant, Machine Vision **Sep. 2025 – Jan. 2026**
Department of Interdisciplinary Sciences and Technologies, University of Tehran
Presented by Dr. Hanieh Naderi

- Designed problems for assignments and exams
- Graded the course assignments

TEACHING EXPERIENCE

Teaching Assistant, Automatic Control **Sep. 2023 – Jan. 2024**
Mechanical Engineering Department, K. N. Toosi University of Technology
Presented by Dr. Naser Naserifar

- Conducted exercise-solving classes
- Produced educational materials

Teaching Assistant, Computer Programming **Feb. 2022 – Jun. 2022**
Mechanical Engineering Department, K. N. Toosi University of Technology
Presented by Dr. Farschad Torabi

- Conducted exercise-solving classes focused on programming in C++ using the Qt framework
- Produced educational materials

TECHNICAL AND RESEARCH EXPERIENCE

- **Physics-informed Data-driven Modeling of a Two-DoF Robot Arm** [GitHub]
  Utilized JAX for an optimized implementation of Lagrangian Neural Networks. This model was used for learning the dynamics of a two-DoF robot arm.
- **Simulation and Control of a 5RP Robot Arm** [GitHub]
  Design, simulation, and implementation of various control techniques on a 5RP robotic manipulator.
- **Design of a Controller for a Robotic Bicycle Using Intelligent Control Techniques** [GitHub]
  Designed linear and nonlinear controllers for a robotic bicycle using the DDPG algorithm. The robot was simulated in MuJoCo.
- **Mobile Robot Control System (teamwork)** [GitHub]
  Designed a PID-based control system for a physical differential-drive mobile robot. A GUI application served as a control panel. The user was able to draw the desired path of the robot in the control panel.
- **Passive Steering Wheel (teamwork)** [GitHub]
  Designed a CNN model capable of estimating the steering angle from an input image of a steering wheel.

PUBLICATIONS

- **Gray-Box Computed Torque Control for Differential-Drive Mobile Robot Tracking** [arXiv] [GitHub]
- **Optimal Intelligent Control of Robotic Manipulators with Optimal Reliance on Physics Information** [TechRxiv]
- **LiteMish: A Computationally Efficient and Smooth Algebraic Alternative to Mish** [TechRxiv][GitHub]

TECHNICAL SKILLS

- Programming languages and mathematical packages: C, C++, MATLAB, Python
- Software packages: Git, Proteus, LTSpice, STM32CubeMX, Keil uVision
- Software libraries and frameworks: Keras, TensorFlow, JAX, OpenCV, MuJoCo, Qt (GUI development)
- Computer aided design: SOLIDWORKS
- Development boards: STM32, ESP32, Arduino
- Operating systems: Windows, Linux
- Hardware skills: Soldering
- Other: Deep RL, Simulink, Network Programming, LaTeX

COURSES STUDIED

Course Name	Grade (out of 20)
Advanced Robotics	19.83
Machine Vision	20.00
Optimal Control	18.75
Modern Control	18.80
Fundamentals of Electrical Engineering I	20.00
Fundamentals of Electrical Engineering II	19.50
Neural Networks	19.00
Measurement and Control Systems	18.50
Engineering Drawing II	19.60
Introduction to Mechatronics	18.00

CURRENT RESEARCH

Developing a novel nonlinear control technique to enforce robust asymptotic stability of uncertain dynamical systems controlled by reinforcement learning agents.

HONORS AND AWARDS

- **Ranked within the top 10% of mechanical engineering students from the same year of entrance**
  Issued by K. N. Toosi University of Technology • Sep. 2024
- **Ranked within the top 1.5% in the Iranian University Entrance Exam**
  Issued by National Organization of Educational Testing • Sep. 2020

LANGUAGES

- **English**
  Proficient
  TOEFL iBT Score: 90
- **Persian**
  Native language

REFERENCES

Khalil Alipour, Associate Professor
College of Interdisciplinary Science and Technology
University of Tehran
(+98 21) 8609 - 3062, k.alipour@ut.ac.ir

Majid Sorouri, Assistant Professor
Department of Electronic Engineering
Maynooth University
majid.sorouri@mu.ie

Farschad Torabi, Associate Professor
Department of Mechanical Engineering
K. N. Toosi University of Technology
(+98 21) 8406 - 3245, ftorabi@kntu.ac.ir